Water Quality Exam Packet

2020 Golden Gate Science Olympiad Invitational

Time limit: 50 minutes

- ***Do not open this exam until you are told to do so***
- ***Write all answers directly on the exam***
- ***Write your team number on every page of the exam***
- ***You may take apart the exam, but you must staple it back together afterward***
- ***The space for recording your salinity measurement is located on page 20***
- ***You do not need to show work to receive full credit for calculations, but it may help you get partial credit for an incorrect answer***
- ***Use correct significant figures when making calculations***
- ***Tiebreakers are questions 24, 25, 28, 33, 68, and 74***

Names: _____

Team Number: _____

School: _____

Score: / 119

Rank:

Team Number: _____

Part I: Marine and Estuary Ecology (37 points)

1. At which oceanic depth would you be **least** likely to find algal blooms? (1 point)
    A. At the surface
    B. 25m
    C. 100m
    D. 250m
    E. Algae does not grow in the ocean

2. Which of the following is a valid reason for the importance of the phosphorus cycle? Select all that apply. (1 point)
    A. Phosphates are a component of DNA
    B. Phosphates are a component of RNA
    C. Phosphates help filter some harmful pathogens through fish gills
    D. Phosphates help regulate the pH of water because they are strong bases.
    E. None of the above

3. Which of the following describes species richness? (1 point)
    A. The total population of a given species in a given community or region.
    B. The number of different species in a given community or region
    C. The ratio of the number of species in a region to the area of that region
    D. The ratio of the population of a species to the biomass of a region
    E. None of the above

4. Which of the following can be used to disinfect wastewater during tertiary treatment? Select all that apply. (1 point)
    A. Ozone
    B. Chlorine
    C. Ultraviolet light
    D. Monochloramine
    E. Sodium hypochlorite

5. Which of the following filter cloth fibers can operate above 325°F (163°C) and would be most effective for handling wastewater with a high concentration of acids? (1 point)
    A. Glass
    B. Asbestos
    C. Teflon
    D. Cotton
    E. Dynel

(For grader's use only) Points: _____

6. Which of the following statements about slow sand filters is false? Select all that apply. (1 point)
    A. Slow sand filters use a biofilm which adsorbs organic material
    B. Slow sand filters are especially effective in treating high turbidity water
    C. Slow sand filters do not require electricity to operate
    D. Slow sand filters do not require chemicals to operate
    E. Slow sand filters work at a constant flow rate

7. Consider a community consisting of predatory reef fish, crown-of-thorns starfish, coral, and algae in the Pacific Ocean. Suppose an increase in the fishing of predatory reef fish causes the population of predatory reef fish to decline significantly. What is the most likely effect of this change? (1 point)
    A. An increase of the local concentration of phosphates
    B. A decrease of the algae population
    C. A decrease of the coral population
    D. A decrease in the crown-of-thorns starfish population
    E. The predatory reef fish population will recover too quickly for any effects to take place

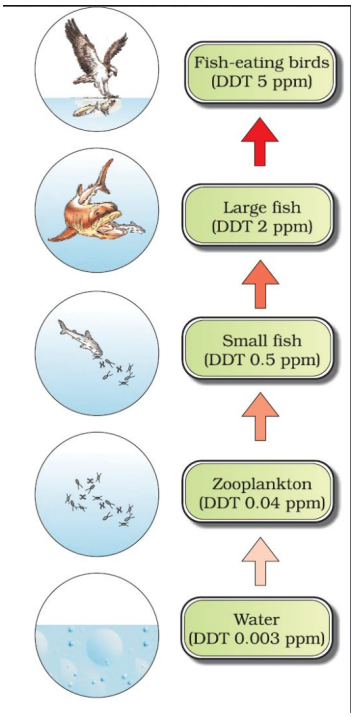
8. Which of the following terms best describes phenomenon occurring in question 7? (1 point)
    A. Biomagnification
    B. Trophic cascade
    C. Succession
    D. Turnover
    E. Bioaccumulation

9. Approximately what percent of the total global primary production is contributed by phytoplankton? (1 point)
    A. 0.01
    B. 0.1%
    C. 1%
    D. 50%
    E. 80%

Team Number: _____

10. Which of the following is the best definition of compensation depth? (1 point)
    A. The depth at which light intensity reaches a level at which oxygen evolved from a photosynthesizing organism equals that consumed by its respiration
    B. The depth at which the ocean's temperature decreases at the greatest rate
    C. The depth at which predator and prey species are balanced such that there is no net change in their populations over time
    D. The depth at which the amount of subducting tectonic matter equals the formation of new tectonic matter
    E. The depth at which no light is able to reach

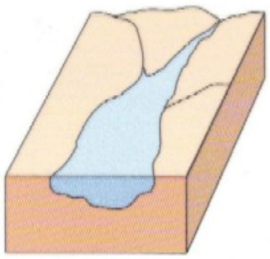
11. The diagram below best illustrates which ecological concept? (1 point)



    A. Bioaccumulation
    B. Biomagnification
    C. Bioconcentration
    D. Density dependent growth
    E. Competitive exclusion

(For grader's use only) Points: _____

12. Classify the pictured estuary. (1 point)



A. Coastal plain/drowned river valley
B. Bar-built
C. Fjord
D. Tectonic
E. None of the above

13. How would decreasing the volume of an estuary affect the residence time of water? (1 point)
    A. Increase
    B. Decrease
    C. No effect
    D. Indeterminate

14. Which of the following organisms would you be **least** likely to find in the abyssopelagic zone? (1 point)
    A. Bluefin Tuna
    B. Anglerfish
    C. Gulper eel
    D. Tripod fish

15. Suppose development of residential areas near the Chesapeake bay has increased soil erosion in the area. What effect would this most likely have on the local filter feeders? (1 point)
    A. It would increase their food supply, increasing their population exponentially
    B. It would decrease their food supply
    C. It would reduce filtration rates
    D. They would be forced to resort to predation to survive
    E. It would have no effect

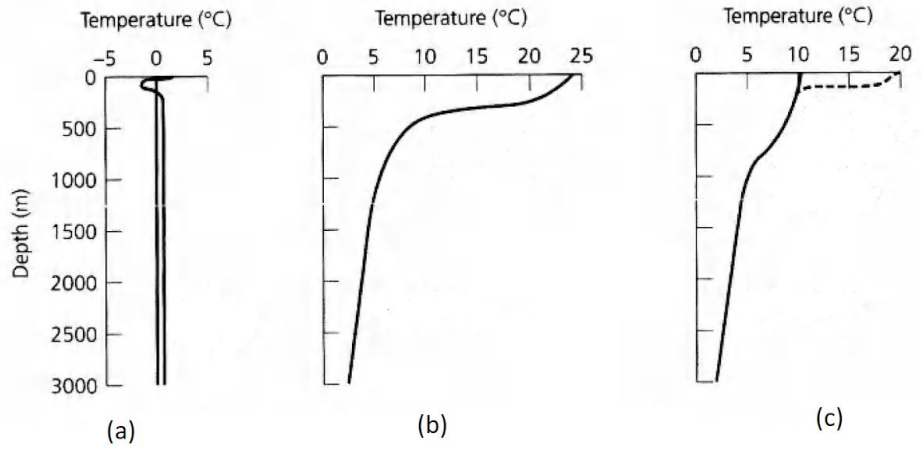
Use images 1, 2, and 3 to answer questions 16, 17, and 18

16. The images display habitat suitability maps for sperm whales (image 1), pilot whales (image 2), and Risso's dolphins (image 3) in the northwestern Mediterranean based on ecological niche factor analysis. 100 indicates the most habitat suitability, and 0 the least. Each of these species prey exclusively or preferentially on cephalopods. The data presented in these maps best illustrates which of the following concepts? (1 point)
    A. Density-dependent growth
    B. Fundamental niche
    C. Competitive exclusion
    D. Intraspecific competition
    E. Resource partitioning

17. Fill in the blank. The answer to question 16 reduces _____ _____ [2 words], allowing the three species to coexist. (1 point)

18. Fill in the blank. Each map represents the given species' _____ _____ [2 words], which is the reduced breadth of its resources in the presence of another competing species. (1 point)

19. Three graphs of thermoclines at different latitudes are shown below. Match each graph to its corresponding latitude by writing the letters in the blanks. (3 points)



Low latitudes _____

Mid-latitudes _____

High latitudes _____

(For grader's use only) Points: _____

20. Give one reason why the freezing point of seawater is below 0 °C. (1 point)

21. Suppose the current population of blue crabs in the Chesapeake Bay is 300 million and the carrying capacity is 650 million. Using the logistic growth model with r = 0.089, determine the population of blue crabs in the Chesapeake Bay in 3 years. (2 points)

22. Determine the maximum sustainable yield of blue crabs using the information in question 21. (1 point)

23. Briefly describe the process by which activated carbon filters drinking water (1 point).

(For grader's use only) Points: _____

24. (Tiebreaker 1) Iron fertilization involves dumping large amounts of soluble iron onto the surface of the ocean. Explain how iron fertilization can help slow climate change. (2 points)

25. (Tiebreaker 2) Explain why phytoplankton have a self-imposed upper bound on their population size (hint: think about turbidity) (3 points).

26. What does the residence time of water measure? (1 point)

27. Many marine vertebrates cope with their saline environments by passing water from high salt concentration to low salt concentration between permeable membranes and excreting excess salt through their skin and gills. They take in water through drinking to maintain the water balance. What is the name for this process? (1 point)

28. (Tiebreaker 3) Blue crabs feed on several types of bivalve mollusks, each having different defense mechanisms. Mussels, for example attach themselves to rocks in open view, but are heavily armored and can close their shell valves completely. Clams on the other hand, bury themselves in the sand and mud and rely on hiding themselves to avoid being eaten. They are harder to find, but compared to mussels, they are easier for blue crabs to eat.

    (a) What type of functional response is exhibited between blue crabs and **clams**? (1 point)

    (b) What is the main factor that determines the type of functional response between blue crabs and **clams**? (1 point)

    (c) What type of functional response is exhibited between blue crabs and **mussels**? (1 point)

    (d) What is the main factor that determines the type of functional response between blue crabs and **mussels**? (1 point)

Part II: Coral Reef Macroflora and Fauna Identification (35 points)

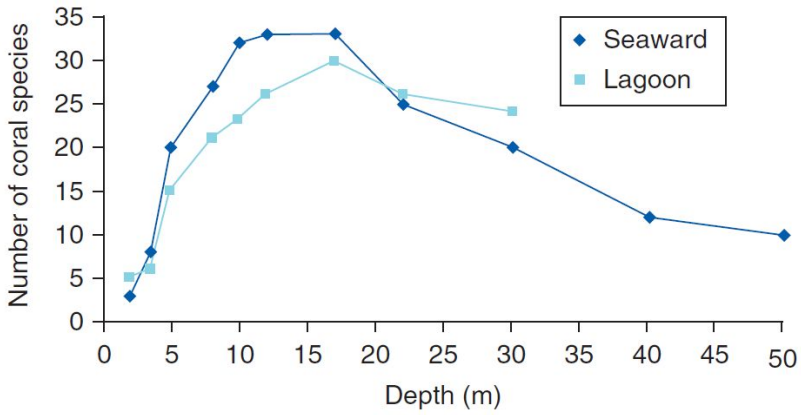
29. Which of the following are characteristics of a fringing reef? Select all that apply. (1 point)
    A. Circular
    B. Separated from shore by deep lagoon
    C. Shallow/no backreef zone
    D. Located far from the shore
    E. Can be found within atolls

30. Which of the following statements best describes the relationship between coral polyps and zooxanthellae? (1 point)
    A. Zooxanthellae contain toxins which serve as a defense mechanism for coral against predators while the coral shelters the zooxanthellae.
    B. Zooxanthellae provide camouflage for coral while the coral shelters the zooxanthellae.
    C. Zooxanthellae consume the coral for nutrients.
    D. Coral takes in inorganic nitrogen from the water and fixes it into organic nitrogen for zooxanthellae.
    E. Coral shelters zooxanthellae and provides it with CO_2 while zooxanthellae provides organic nutrients to the coral.

31. Which of the following pose threats to the health of the Great Barrier Reef? Select all that apply. (1 point)
    A. Climate change
    B. Crown-of-thorns starfish
    C. Overfishing
    D. Eutrophication
    E. Storms

32. Describe how the removal of urchins and herbivorous fish from a reef system would affect the health of the reef. (2 points)

33. (Tiebreaker 4) The graph below shows the number of coral species vs. depth in the Chagos Atoll in the Indian Ocean.



Despite the fact that coral species need significant amounts of light to thrive, the maximum number of coral species occurs between a depth of 10 and 20m rather than 1 and 10m. Give one likely reason for this trend. (2 points)

34. How does a coral become bleached? (1 point)

35. How do branching corals outcompete other coral species? (1 point)

Use image 4 to answer questions 36-39

36. Give the common name of the organism shown in the image. (1 point)

37. Which of the following interactions does this organism exhibit with other reef species when it is feeding? (1 point)
    A. Parasitism
    B. Commensalism
    C. Amensalism
    D. Competition
    E. Mutualism

38. How does this organism signal that it is ready to feed? (1 point)

39. Which sex is physically larger in this organism's species, male or female? What term is used to describe this difference in characteristics between sexes? (2 points)

Use image 5 to answer questions 40-43

40. Give the common name of the organism shown in the image. (1 point)

41. What role does this organism play in the bioturbation of the sea bed? (2 points)

42. In what way is this organism detrimental to reef health? (1 point)

43. Which of the following are ways in which this organism defends itself from predators? Select all that apply. (1 point)
    A. It utilizes its high maneuverability to evade predators
    B. It expels sticky tubules which entangle predators
    C. It opens the ion channels within its body to create a potential difference and electrocute predators
    D. Its body contains various toxins which are harmful to predators
    E. None of the above

Use image 6 to answer questions 44-46

44. Give the common name of the organism shown in the image. (1 point)

45. How long does this organism take to reach sexual maturity? (1 point)
    A. 1-3 years
    B. 4-6 years
    C. 7-9 years
    D. 10-12 years
    E. 14-16 years

46. Which of the following conditions does this organism exhibit? (1 point)
    A. Simultaneous hermaphroditism
    B. Pseudohermaphroditism
    C. Protandry
    D. Protogyny
    E. Bidirectional Sex Changers

Use image 7 to answer questions 47-49

47. Give the common name of the organism shown in the image. (1 point)

48. What is the planktonic larval duration of this organism? (1 point)

49. What does this organism's diet mostly consist of? (1 point)

Use image 8 to answer questions 50-52

50. Give the common name of the organism shown in the image. (1 point)

51. Describe the caridoid escape reaction exhibited by this organism. (2 points)

52. Which of the following is **not** a predator of this organism? (1 point)
    A. Cod
    B. Flounder
    C. Eels
    D. Tuna
    E. Seals

Use image 9 to answer questions 53-55

53. Give the common name of the organism shown in the image. (1 point)

54. Explain the difference between active and passive filter feeders. Is this organism an active or passive filter feeder? (3 points)

55. How does this organism antagonize corals while competing for reef space? (2 points)

Part III: Water Monitoring and Analysis (35 points)

56. A hydrometer can measure salinity based on which of the following concepts? (1 point)
    A. Conductivity
    B. Resistance
    C. Capacitance
    D. Buoyancy
    E. Temperature

Use image 10 to answer question 57

57. Which two quantities do the curves on the graph most likely represent? (1 point)
    A. Dissolved oxygen and temperature
    B. pH and chlorophyll
    C. pH and turbidity
    D. Total solids and turbidity
    E. Salinity and nitrate concentration

58. If the temperature of a body of water increases and its pH decreases, how would you expect the unionized ammonia concentration to change? (1 point)
    A. Increase
    B. Decrease
    C. Indeterminate

59. A sample of water has a pH of 3.0. A doubling of the pH from 3.0 to 6.0 would change the acidity by a factor of (1 point)
    A. 2
    B. 0.5
    C. 1000
    D. 0.01
    E. 0.001

60. Which of the following best describes the relationship between the depth of the aragonite saturation horizon and the age of water? (1 point)
    A. Direct
    B. Inverse
    C. No relationship

61. A scientist passes a 0.50 L sample of water through a filter. The scientist weighs the filter paper and finds that the combined mass of the solids and the filter paper is 650.0 mg. Next, the scientist boils off the water in the remaining filtrate. They find that the mass of the solids from the filtrate and the container holding the filtrate combined is 30.12 g. If the filter paper's mass is 100.0 mg and the container holding the filtrate is 30.00 g, determine:

    (a) The concentration of total suspended solids in g/L. (2 points)

    (b) The concentration of total dissolved solids in g/L. (2 points)

62. Suppose the attenuation coefficient of the San Francisco Bay at a particular point in time is 0.40 m^{-1}. Determine the secchi depth in meters of the San Francisco Bay. (2 points)

63. Considering your result in question 62, use Carlson's Trophic State Index to classify the San Francisco Bay as oligotrophic, mesotrophic, eutrophic, or hypereutrophic. (1 point)

Use image 11 (Chlorophyll in South San Francisco Bay) to answer question 64

64. What is the most likely cause of the annual spikes in the graph? (2 points)

65. Describe the conventional laboratory procedure for determining BOD_5 (biochemical oxygen demand). (3 points)

66. Suppose a particular body of water has an aragonite saturation state of 0.86. How would you expect the pH of this water to change in the immediate future? (1 point)

Use images 12, 13, 14, and 15 to answer question 67

67. Match each spatial distribution of aragonite saturation state (labeled A-D) to the corresponding oceanic depth. (2 points)

50 meters: _____ 100 meters: _____ 200 meters: _____ 500 meters: _____

Use image 16 to answer question 68

68. (Tiebreaker 5) The image shows a diagram of a water quality testing device. What is the name of this device and what does it measure? (2 points)

69. The table below shows various measurements taken from a body of water.

Temperature	28℃
Salinity	35 ppt
pH	4.5
Dissolved Oxygen	23.3%
Turbidity	19 NTU
Phosphates	0.01 mg/L

Is this body of water suitable for supporting diverse aquatic life? Justify your answer. (2 points)

70. Briefly describe how an increase of surface temperature of a body of water would affect each of the following.

(a) Rate of evaporation of the water (1 point)

(b) Concentration of dissolved solids and nutrients (1 point)

(c) Turbidity (1 point)

71. A 20.0 mL sample of wastewater is placed into a 300 mL BOD bottle. The rest of the bottle is filled with aerated dilution water. The initial dissolved oxygen was 9.70 mg/L and after 5 days the dissolved oxygen was 1.20 mg/L.
    (a) What is the BOD_5 of the water sample in mg/L? (2 points)

    (b) Would this water be safe to discharge to a nearby river? Justify your answer by interpreting your result in part (a). (2 points)

72. Warm water holds (less / more) oxygen than cold water (circle one). Why is this? (2 points)

73. Why does seawater tend to have lower turbidity than freshwater? (2 points)

Part IV: Salinometer Testing (12 points)

74. (Tiebreaker 6) Record salinity measurement as a percentage here (6 points): _____

Brought salinometer for testing (6 points):
(DO NOT FILL OUT THIS LINE) Event supervisor initials: _____

(For grader's use only) Points: _____